Customer Support Chatbot MLOps and Agentic AI System

Enterprise MLOps & Agentic AI System

Active Model: `v1.0.1`

Framework: `Scikit-Learn + TF-IDF`

Orchestrator: `5-Agent Pipeline`

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- Batch Processing & CSV
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MLOps & Model Governance

Experiment tracking, versioned model registry, confusion matrices, and latency benchmarks.

Active Model Version	Category Macro-F1	Urgency Macro-F1	Inference Latency (p95)
v1.0.1	1.0	0.8963	4.98 ms
↑ Production Active	↑ Accuracy: 100.0%	↑ Accuracy: 90.1%	↑ p50: 2.45 ms

Confusion Matrix: Category Classification



Versioned Model Registry

	version	is_active	created_at	macro_f1
0	v1.0.0	☐	2026-10-12 15:51:48	1
1	v1.0.1	☑	2026-10-13 09:27:51	1

Local Experiment Run History

	version	timestamp	train_samples	test_samples	cat_f1	urg_f1	p50_latency_ms
0	v1.0.0	2026-10-12 15:47:53	560	140	1	0.9107	0.67
1	v1.0.0	2026-10-12 15:51:48	560	140	1	0.9013	0.67
2	v1.0.1	2026-10-13 09:27:51	563	141	1	0.8963	2.45

QLoRA Fine-Tuning (Low-Rank Parameter-Efficient Adaptation)

Fine-tune instruction-aligned adapter heads with frozen backbone parameters and 4-bit/low-rank factorization.

Adapter Rank (r)

16

↑ alpha = 32.0

Target Modules

q_proj, v_proj

↑ LoRALinear Injection

Trainable Parameter Ratio

1.62%

↑ 98.38% Backbone Frozen